THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** $(-5, -2)$	Reflection in the y-axis changes the sign of the abscissa.
2	**(c)** $3x + 4$	Linear means degree exactly 1.
3	**(d)** 3	It is the larger of the two powers.
4	**(b)** $4x^2 + 12xy + 9y^2$	$(a+b)^2 = a^2 + 2ab + b^2$
5	**(a)** 5	Apply the distance formula.
6	**(c)** -10	Put x in $p(x) = 0$.
7	**(b)** $\frac{3}{11}$	Multiply by 100 and subtract.
8	**(a)** $9x^2 - 12xy + 4y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
9	**(a)** $(2, 2)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
10	**(d)** -1	Substitute $x = -k$.
11	**(b)** $\sqrt{5}$	$2^2 < 5 < 9$
12	**(b)** $x^2 + 4x - 21$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
13	**(b)** $(6, 0)$	Ordinate is 0 on the x-axis.
14	**(b)** 7	Put $x = 2$.
15	**(d)** irrational	Use the rules for sums and products of rational and irrational numbers.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$PQ = \sqrt{(3)^2 + (4)^2}$ (1); $PQ = 5$ units (1).	1+1
17	$3x - 18 = 0$ gives $x = 6$ (1); The graph meets the x-axis at $(6, 0)$ (1).	1+1
18	$x = 0.\overline{3}$, $10x - x = 3$ (1); $x = \frac{1}{3}$ (1).	1+1
19	Use the square identity (1); $= 9x^2 + 30xy + 25y^2$ (1).	1+1
20	$\left(\frac{-3+(9)}{2}, \frac{7+(1)}{2}\right)$ (1); Midpoint $= (3, 4)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 64$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
22	$x = -1$: $y = 1$; $x = 0$: $y = 3$ (1); $x = 1$: $y = 5$; $x = 2$: $y = 7$ (1); Slope $= 2$, y-intercept $= 3$ (1).	1+1+1
23	$x = 0.\overline{36}$, $100x = 36.\overline{36}$ (1); $99x = 36$ (1); $x = \frac{4}{11}$ (1).	1+1+1
24	Use the square identity (1); $= 4x^2 + 12xy + 9y^2$ (1); LHS $= (2+3)^2 = 25$, RHS $= 25$ (1).	1+1+1
	OR: Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x-(-1))^2 + 16 = (x-5)^2 + 16$ (1); $x = 2$, point $(2, 0)$ (1).	1+1+1
25	Substitute: $11 = 6 + k$ (1); $k = 5$ (1); Line: $y = 2x + 5$ (1).	1+1+1
	OR: $x = 0.16\overline{6}$; $10x = 1.6\overline{6}$ and $100x = 16.6\overline{6}$ (1); $90x = 15$ (1); $x = \frac{1}{6}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$AB^2 = 17$, $DC^2 = 17$: $AB = DC$ (1); $BC^2 = 13$, $AD^2 = 13$: $BC = AD$ (1); Midpoint of $AC = (3, 2)$ (1); Midpoint of $BD = (3, 2)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
	OR: $C(x) = 12x + 60$ (1); $C(5) = 120$ (1); $12x + 60 = 168$ (1); $x = 9$ (1); Hence the number of notebooks is 9 (1).	1+1+1+1+1
27	Remainders repeat after 6 steps (1); $\frac{1}{7} = 0.\overline{142857}$ (1); Block length $= 6$ (1); Non-terminating and repeating: rational (2).	1+1+1+2

	OR: Use $(a-b)^2 = a^2 - 2ab + b^2$ (1); $a = 2x$, $b = 3y$ (1); Squares: $4x^2 + 9y^2$ (1); Cross term: $-12xy$ (1); $= 4x^2 - 12xy + 9y^2$; at $x = 3$, $y = 1$ the value is 9 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 10609	Write $103 = 100 + 3$
2	**(d)** Quadrant II	Use the sign rules for the quadrants.
3	**(b)** $(4, 0)$	Put $y = 0$.
4	**(d)** $\frac{1}{6}$	Use the repeating-decimal method.
5	**(b)** $(x + 4)$	$a^2 + 2ab + b^2 = (a + b)^2$
6	**(b)** 7	The distance from the x-axis is $\lvert y \rvert$.
7	**(c)** 7	Find each zero.
8	**(d)** 8	Use Brahmagupta's rules.
9	**(d)** $(x + 2)(x + 5)$	Find two numbers with the given sum and product.
10	**(a)** $(8, 11)$	$B = 2M - A$ coordinatewise.
11	**(c)** $6n - 5$	Use $a_n = a_1 + (n - 1)d$.
12	**(d)** $\frac{7}{16}$	Terminates when the denominator has only 2 and 5 as prime factors.
13	**(c)** 53	$x^2 + y^2 = (x + y)^2 - 2xy$.
14	**(d)** $(3, 8)$	Reflection in the y-axis changes the sign of the abscissa.
15	**(b)** 250	Fare = fixed + rate × time.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$p(2) = 11$ (1); $p(-2) = -5$ (1).	1+1
17	Divide 1 by 6 (1); $= 0.1\overline{6}$ (1).	1+1
18	$104 = 100 + 4$ (1); $104^2 = 10816$ (1).	1+1
19	$B = (2 \cdot 4 - (1),\ 2 \cdot 6 - (2))$ (1); $B = (7, 10)$ (1).	1+1
20	Slope $= 8$, intercept $= 3$ (1); Cuts y-axis at $(0, 3)$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Need two numbers with sum 8 and product 12 (1); They are 2 and 6 (1); $= (x + 2)(x + 6)$ (1).	1+1+1
22	$D = (1, 29)$ (1); $AC = \sqrt{21^2 + 28^2}$ (1); $AC = 35$ units (1).	1+1+1
23	$p + q = 5x + 10$ (1); Put $5x + 10 = 0$ (1); Zero $x = -2$ (1).	1+1+1
24	Write with denominator 12: $\frac{4}{12}$ and $\frac{8}{12}$ (1); Numerators 5, 6, 7 lie between (1); Numbers: $\frac{5}{12}$, $\frac{6}{12}$, $\frac{7}{12}$ (1).	1+1+1
	OR: $103 \times 107 = (100 + 3)(100 + 7)$ (1); $= 100^2 + (10)(100) + (21)$ (1); $= 11021$ (1).	1+1+1
25	$AB^2 = 25$ (1); $BC^2 = 25$, $CA^2 = 64$ (1); Two sides are equal, so the triangle is isosceles (1).	1+1+1
	OR: $x = -1$: $y = -4$; $x = 0$: $y = -1$ (1); $x = 1$: $y = 2$; $x = 2$: $y = 5$ (1); Slope $= 3$, y-intercept $= -1$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$AB = \sqrt{225} = 15$ (1); $BC = \sqrt{144} = 12$ (1); $CA = \sqrt{81} = 9$ (1); Perimeter $= 36$ units (2).	1+1+1+2
	OR: $y(0) = -12$, $y(1) = -8$, $y(2) = -4$ (1); Crosses x-axis: $x = 3$ (1); Point $(3, 0)$ (1); Slope $= 4$ (1); Slope is positive: the graph rises (1).	1+1+1+1+1
27	First $= 5^2 - 3 = 22$ (rational) (1); Second $= 25 + 3 + 10\sqrt{3}$ (1); $= 28 + 10\sqrt{3}$ (1); Irrational (contains a surd) (1); Hence only the first result is rational (1).	1+1+1+1+1
	OR: Use $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ (1); $a = x$, $b = 2$ (1); $3a^2b = 6x^2$, $3ab^2 = 12x$ (1); $= x^3 + 6x^2 + 12x + 8$ (1); Terms checked (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 1–4: **Coordinates, Linear Polynomials, World of Numbers, Algebraic Identities** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	It is the larger of the two powers.
2	**(d)** $a^3 + 3a^2b + 3ab^2 + b^3$	Standard cube identity.
3	**(a)** 13	Apply the distance formula.
4	**(a)** $5 - 2x$	Linear means degree exactly 1.
5	**(b)** $\frac{5}{11}$	Multiply by 100 and subtract.
6	**(d)** $9x^2 + 24xy + 16y^2$	$(a+b)^2 = a^2 + 2ab + b^2$
7	**(c)** $(-2, 4)$	Use $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$.
8	**(c)** -12	Put x in $p(x) = 0$.
9	**(d)** $\sqrt{11}$	$3^2 < 11 < 16$
10	**(d)** $4x^2 - 20xy + 25y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
11	**(c)** $(-4, 0)$	Ordinate is 0 on the x-axis.
12	**(a)** -1	Substitute $x = -k$.
13	**(d)** irrational	Use the rules for sums and products of rational and irrational numbers.
14	**(a)** $x^2 + 3x - 40$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
15	**(c)** Quadrant IV	Use the sign rules for the quadrants.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Average: $\frac{3+4}{2}$ (1); $= \frac{7}{2}$ (1).	1+1
17	Use $(x+a)(x+b) = x^2 + (a+b)x + ab$ (1); $= x^2 + 13x + 40$ (1).	1+1
18	$PQ = \sqrt{(8)^2 + (-15)^2}$ (1); $PQ = 17$ units (1).	1+1
19	$5x + 20 = 0$ gives $x = -4$ (1); The graph meets the x-axis at $(-4, 0)$ (1).	1+1
20	$x = 0.\overline{6}$, $10x - x = 6$ (1); $x = \frac{2}{3}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$x = 0.\overline{15}$, $100x = 15.\overline{15}$ (1); $99x = 15$ (1); $x = \frac{5}{33}$ (1).	1+1+1
22	Use the square identity (1); $= 9x^2 - 24xy + 16y^2$ (1); LHS $= (3-4)^2 = 1$, RHS $= 1$ (1).	1+1+1
23	Let the point be $(x, 0)$; set $PA^2 = PB^2$ (1); $(x-(-5))^2 + 4 = (x-3)^2 + 4$ (1); $x = -1$, point $(-1, 0)$ (1).	1+1+1
24	Substitute: $4 = 6 + k$ (1); $k = -2$ (1); Line: $y = 3x - 2$ (1).	1+1+1
	OR: $x = 0.2\overline{7}$; $10x = 2.\overline{7}$ and $100x = 27.\overline{7}$ (1); $90x = 25$ (1); $x = \frac{5}{18}$ (1).	1+1+1
25	Need two numbers with sum -7 and product 10 (1); They are -5 and -2 (1); $= (x-5)(x-2)$ (1).	1+1+1
	OR: $D = (-2, 35)$ (1); $AC = \sqrt{24^2 + 32^2}$ (1); $AC = 40$ units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$AB^2 = 17$, $DC^2 = 17$: $AB = DC$ (1); $BC^2 = 13$, $AD^2 = 13$: $BC = AD$ (1); Midpoint of $AC = (4, 3)$ (1); Midpoint of $BD = (4, 3)$ (1); Opposite sides equal and diagonals bisect each other: parallelogram (1).	1+1+1+1+1
	OR: $C(x) = 15x + 45$ (1); $C(4) = 105$ (1); $15x + 45 = 150$ (1); $x = 7$ (1); Hence the number of notebooks is 7 (1).	1+1+1+1+1
27	Remainders repeat after 6 steps (1); $\frac{1}{13} = 0.\overline{076923}$ (1); Block length $= 6$ (1); Non-terminating and repeating: rational (2).	1+1+1+2
	OR: Use $(a-b)^2 = a^2 - 2ab + b^2$ (1); $a = 3x$, $b = y$ (1); Squares: $9x^2 + y^2$ (1); Cross term: $-6xy$ (1); $= 9x^2 - 6xy + y^2$; at $x = 2$, $y = 4$ the value is 4 (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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